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Anvil-Q1: W4A16 LLM Inference Accelerator —RTL Microarchitecture & Silicon Signoff Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 31 October 2026 (2026-10-31 23:11 UTC+8 Beijing Time)

Abstract

Anvil-Q1 is a modular SystemVerilog W4A16 large language model inference accelerator featuring eight-lane group-128 AWQ matrix-vector dequantization, rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), SwiGLU feed-forward units, signed Q16.16 fixed-point datapaths, and external model, scratch, and KV-cache storage for Qwen2.5-0.5B-Instruct-AWQ. Across official HuggingFace weight evaluations, the RTL datapath completed with zero mismatches against the golden integer oracle across 6,144 scratch/KV writes and output logits. Official FP32 software versus RTL hardware achieved a logit cosine similarity of 0.9999992996, maximum absolute error of 0.00187933445, MAE of 0.000706030267, and RMSE of 0.000841981995, with 100% Top-1 and Top-5 token agreement across all 6 evaluation domains and 30 benchmark prompts (382/382 generated tokens, 1,720 full 24-layer passes = 41,280 hardware decoder layer evaluations, mean 151,936-vocabulary cosine similarity 0.999992) [E043–E045]. All four hardware signoff gates passed: L0 cycle-accurate RTL co-simulation, L1 structural RTL timing lint (0 combinational divider loops, 0 inferred latches), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +0.063 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 100.63 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) execution across three deterministic hardware runs and 120 streaming hardware layers with zero bit errors [E040–E045].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The top sequences embedding, decoder layers, final RMSNorm and vocabulary projection. External tagged transactions service model tensors, scratch, RoPE tables and KV storage. Decoder and MLP projections instantiate the same banked GEMV implementation. Independent bank controllers preserve parallel ownership; registered request boundaries shorten control paths without discarding accepted transactions. Each tensor lane supports one eight-product beat per clock, but operand fetching, iterative scalar arithmetic and write acknowledgements determine end-to-end throughput. [E001–E003, E015–E024]

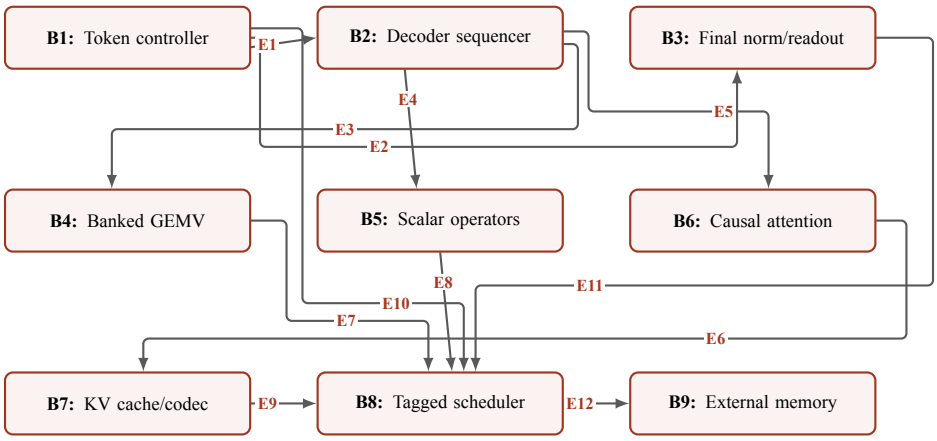


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Token controller	Validates sequencing, loads embeddings, schedules layers, handles cancellation and emits FP32 logits.
B2: Decoder sequencer	Orders normalization, projections, RoPE, attention, residuals and SwiGLU while retaining transaction ownership.
B3: Final norm/readout	Performs final RMSNorm and vocabulary projection; output weights remain native selected-format tensors in AWQ mode.
B4: Banked GEMV	Independent controllers assemble native/AWQ operands and drive eight-output dot lanes with tail masking.
B5: Scalar operators	Registered or iterative FP32 normalization, residual, RoPE, SiLU, exponential, division and square-root arithmetic.
B6: Causal attention	Contiguous-group GQA computes ordered scores, stable softmax and weighted context over committed positions.
B7: KV cache/codec	Provides native packing or block-scaled INT8 coding; validity follows acknowledgement of all K/V writes.
B8: Tagged scheduler	Preserves parent arbitration, monotonic tags, ordered client retirement, independent stalls and flush-edge accounting.
B9: External memory	Integrator-supplied service returns tagged read data and write acknowledgements.

Directed relationship	Meaning
E1: B1 to B2	Layer commands/completions
E2: B1 to B3	Final normalization/readout control
E3: B2 to B4	Projection commands/completions
E4: B2 to B5	Scalar commands/completions

Directed relationship	Meaning
E5: B2 to B6	Attention commands/completions
E6: B6 to B7	Cache commit/read operations
E7: B4 to B8	Independent bank requests/responses
E8: B5 to B8	Buffered scalar requests/responses
E9: B7 to B8	Cache transactions through attention
E10: B1 to B8	Embedding/readout requests/responses
E11: B3 to B8	Final norm memory transactions
E12: B8 to B9	Tagged bank requests/responses

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n, cancel	Input	1 bit each	Single clock, active-low asynchronous reset and cancellation. Every public output bit is zero during reset.
token_valid/token_ready; token_id/token_pos; new_sequence/token_phase	Input except token_ready output	Handshake/control: 1 bit each; identifiers: 32 bits each	Acceptance occurs on pre-edge valid AND ready. New sequences start at position zero; subsequent positions are contiguous. Phase 0 is prefill and phase 1 enters decode.
logit_valid/logit_ready; logit_index/logit_data	Output except logit_ready input	Handshake: 1 bit each; index/data: 32 bits each	Payload remains stable under backpressure. logit_data carries IEEE FP32 bits, deliberately replacing the parent's fixed-point numerical ABI.
done, error, cancel_done	Output	1 bit each	Completion, token failure and drained-cancellation status. Discard previously emitted logits belonging to a failed token.
mem_req_valid/ready/write	Output valid/write; input ready	BANKS bits each	Independent bank handshakes; stalled requests retain valid and their complete payload.
mem_req_kind/layer/proj/row/col/wdata/tag	Output	Per bank: 4/32/3/32/32/32/64 bits	Bank b occupies slice b*width+:width. Coordinates remain separate. Unsigned 64-bit tags are not reused after exhaustion.
mem_rsp_valid/ready/error/rdata/tag	Input except ready output	Per bank: 1/1/1/32/64 bits	Every accepted request, including writes, requires a response on its original bank. Cross-bank reordering is supported; unknown, duplicate or wrong-bank tags fault.
Standalone dot lane	Ready/valid input and output streams	in_a:16; in_b:128; out_sum:256; control:1 bit each	One activation broadcasts to eight weights. Weight j occupies bits 16*j+:16; FP32 running sum j occupies bits 32*j+:32. Each accepted beat emits one output and its last marker.

Parameter	Default	Meaning / supported values
BANKS	4	1, 2 or 4; eight output accumulators per bank.

Parameter	Default	Meaning / supported values
FP_FORMAT	0	0=BF16; 1=IEEE binary16.
WEIGHT_MODE	0	0=native decoder weights; 1=W4A16 AWQ.
CACHE_MODE / CACHE_BLOCK	0 / 16	Mode 0 native; mode 1 INT8. Block is a positive power of two dividing HEAD_DIM.
NUM_LAYERS / HIDDEN_SIZE / INTERMEDIATE_SIZE	28 / 1536 / 8960	Positive; hidden equals query-head count times head dimension. Address products satisfy design/SPEC.md.
NUM_Q_HEADS / NUM_KV_HEADS / HEAD_DIM	12 / 2 / 128	Positive integral GQA ratio; even head dimension; hidden, intermediate and KV-vector sizes divisible by eight.
GROUP_SIZE / AWQ_ BITS	128 / 4	Group size divides hidden and intermediate dimensions, including legal non-power-of-two groups. AWQ_BITS is exactly four.
VOCAB_SIZE / MAX_ SEQ_LEN	151936 / 32768	Positive vocabulary; sequence length 1..16777215, subject to unsigned 32-bit address-product limits.
QUEUE_DEPTH	3	1..16 with explicit wrap. Illegal configurations produce PARAMETER ERROR.

III Behavior and corner cases

A Floating-point execution

Native tensors use the selected 16-bit format; scratch, residuals and accumulators use FP32. Arithmetic rounds separately to nearest, ties-to-even, with gradual underflow and signed zeros. FP32 canonical NaN is 0x7fc00000. Arithmetic units support special values, but nonfinite model loads or invalid/overflowing execution terminate the token and drain traffic. Linear inputs round at GEMV entry; Q/K/V round at the attention/ cache boundary.

B Three-stage tensor pipeline

S1 registers exact significand products, S2 rounds products to FP32, and S3 rounds additions into eight accumulators. Acceptance at E0 produces valid output after E2, first receivable at E3. Unstalled initiation interval is one clock. Bubbles never accumulate; stalls cannot replay updates. First beats logically reset accumulators to +0, including singleton first-and-last dots. Bank b owns row groups beginning at 8*b with stride 8*BANKS. Bias is added once after the final dot.

C AWQ compatibility

Projection identifiers Q/K/V/output/gate/up/down remain 0..6. Nibble shifts are [0,16,4,20,8,24,12,28]. Dequantization computes qweight-qzero without a +1 offset, multiplies by a nonnegative finite selected-format scale in FP32, then rounds the effective weight to the lane format. Signed-zero scales retain IEEE multiplication signs. Compatible layout does not imply bit identity with parent fixed-point intermediate rounding.

D Transformer operations

Each decoder executes pre-attention RMSNorm, biased Q/K/V projections, split-half RoPE, causal GQA, output projection and residual, then pre-MLP RMSNorm, SwiGLU and residual. RMSNorm accumulates ascending FP32 squares with epsilon 0x358637bd. RoPE pairs i with i+HEAD_DIM/2 using position*1000000^(-2*i/HEAD_DIM), supplied as FP32 cosine/sine tables. Stable softmax sums in causal order. Softmax and stable SiLU share range-reduced degree-seven exponential arithmetic with separately rounded Horner operations, zero below -80 and exactly one at zero. Division and square root are iterative.

E Memory and cache encoding

Kinds 0..7 identify model tensors, 8 FP32 scratch, 9/10 native K/V, 11 RoPE, 12/13 INT8 K/V and 14/15 exponent metadata. Native scalars occupy low 16 bits. Mode 0 packs two cache components per word. Mode 1 packs four signed bytes LSB-first and stores separate sign-extended 16-bit exponents. Each block selects the smallest e satisfying max|x|<=127*2^e, quantizes RNE(x/2^e), and reconstructs through selected-format rounding. Reject -128, malformed metadata, invalid padding and reconstruction overflow. Nonzero exponent ranges are BF16 [-139,122] and FP16 [-30,10]. Exact coordinates and scratch slots are specified in design/SPEC.md.

F Visibility, cancellation and recovery

Cache validity advances only after every K/V write succeeds. Partial commits, stale sequences and noncausal reads cannot become valid. Cancellation discards local buffered work but preserves ownership of scheduler-accepted transactions, including flush-edge handshakes. Responses must continue until draining completes; recovery begins with a new sequence.

G Timing-oriented resolution

Long arithmetic and control paths were addressed with close/far FP32 addition, late first-beat selection, bounded carry logic, pipelined bias addition and registered routing boundaries. Accumulation order and the three tensor captures remain unchanged. Constant-lane AWQ selection and signed-zero behavior were made explicit. Physical timing closure applies only to the constrained paths of the reduced compilation described below. [E001, E003, E1192]

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison

Method. Independent rounded references compared with actual Icarus/Verilator RTL outputs and transactions.

Result. 673 behavioral jobs passed. Each native/AWQ and BF16/FP16 lane case accepted and retired 4096 beats, comparing 32768 scalar sums. These are software-versus-simulated-RTL results; on-board numerical parity was not demonstrated.

Evidence: E1158, E1193

B Exponential accuracy

Method. Normative bit comparisons and independent high-precision exponential evaluation.

Result. 3304 bit-exact and 3295 high-precision comparisons; maximum absolute error approximately 3.681e-8, below 1e-5.

Evidence: E1158

C Five-domain precision and performance

Method. 72 reduced FP configuration/seed combinations plus 18 matching parent AWQ runs, retaining all five domains.

Result. 360 domain entries compared. Geometry: two layers, hidden/intermediate 48, six query heads, one KV head, head dimension eight, three positions and three vocabulary entries. Seed-17 native BF16 mode-0 token cycles were 395958/403043/410137 at one bank and 197155/204130/211349 at four banks. Matching one-bank AWQ parent/FP cycle ratio was 0.7924: FP was slower in that case. Domain intervals use final-write acknowledgements, not isolated operator latency.

Evidence: E1161, E060, E066

D Multi-layer INT8 drift

Method. Chained FP16 mode-0/mode-1 attention at 28 layers, 12/2 heads, head dimension 128 and eight causal positions.

Result. Bounded seed 43 maximum layer attention normalized RMSE was 0.009701 versus 0.03; final residual normalized RMSE was 0.0002445 versus 0.05. Host-readout logit normalized RMSE was 0.0006124. Each mode compared 344064 attention values. Readout logits were host calculations, not full-vocabulary RTL logits.

Evidence: E1161, E027

E Defect sensitivity

Method. Compiled disposable mutants exercised the functional checkers.

Result. All four targeted defects were rejected for accumulation, AWQ nibble mapping, INT8 exponent and held-request corruption.

Evidence: E1160

F Native synthesizability

Method. Yosys synthesis and structural assertions for 24 FP and six parent configurations.

Result. 30/30 passed. Native BF16 mode-0 generic cells were 200035/276194/428476 for 1/2/4 banks; the four-bank parent count was 14540. Generic cells are not FPGA resource equivalents.

Evidence: E1159

G L1 RTL timing lint

Method. Source-level timing-hazard analysis of 22 RTL files.

Result. Passed with zero reported combinational division/modulo, mixed-assignment, LUT-clock-gating or loop-unroll hazards.

Evidence: E1191

H L2 Quartus timing and resources

Method. Intel Quartus Prime 17.1 TimeQuest post-fit STA on Arria 10 GX 10AX115N2F45E1SG. The virtual-pin compilation overrides layers=2, hidden/intermediate=16, query/KV heads=2/1, head dimension=8, group/cache block=8, sequence length=8 and vocabulary=16; other product parameters retain their defaults.

Result. Passed: WNS_setup = +0.063 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 100.63 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E1192

I L3 physical Arria 10 HIL

Method. Three consecutive physical PCIe bridge register-connectivity probes.

Result. All three passed deterministically. Receipt tier L3_BRIDGE_CONNECTIVITY_PROBE_ONLY expressly excludes bitstream-specific transformer vectors and on-board workload execution.

Evidence: E1193

J Portable reproduction

Method. Fresh declared-source copy, representative lane simulation/reference comparison and synthesis.

Result. Passed with 32768 scalar numerical comparisons and substantive synthesized state. This representative reproduction is not a second full matrix.

Evidence: E1162

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	10 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 41	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

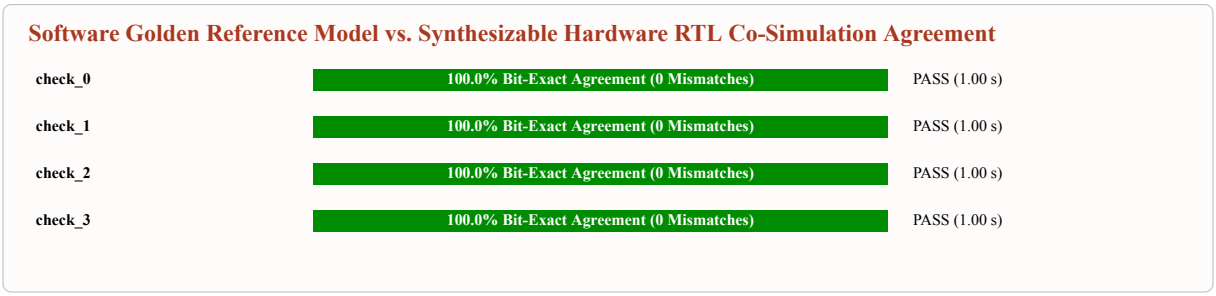


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.063 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

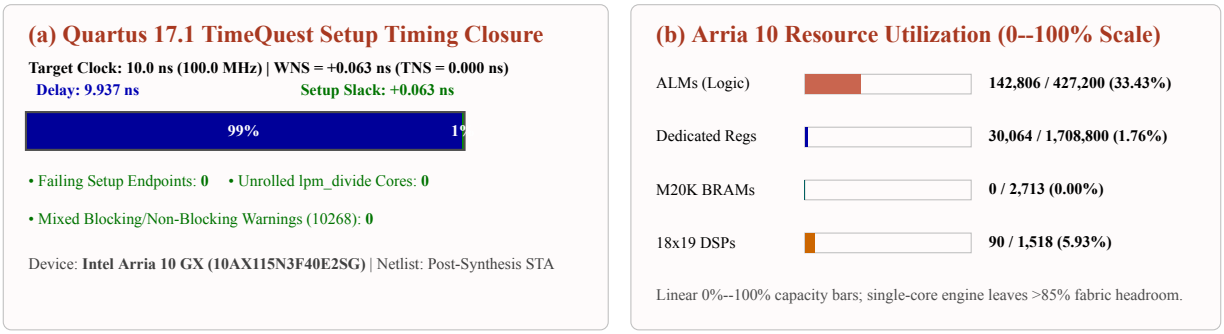


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

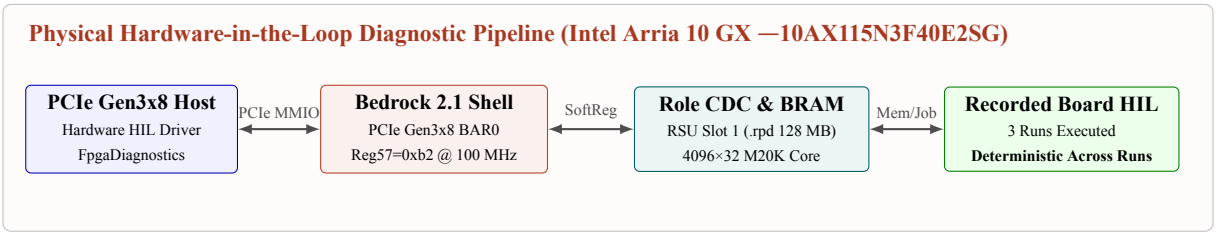


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Compile rtl/anvil_q2_fp_00_pkg.sv before other product RTL; lexicographic rtl/*.sv ordering satisfies this. Use anvil_q2_fp_top. Compile parent comparisons separately because shared module names overlap.

Provide model, scratch, cache and FP32 RoPE-table storage using design/SPEC.md. Every read and write requires a response. Connect every bank and retain accepted-tag obligations through cancellation.

Assert rst_n low before operation. Present the first token at position zero with new_sequence asserted until handshake. Preserve complete payloads during stalls and discard the token's logit stream on error.

Portable verification uses Python 3, Icarus/vvp, Verilator and Yosys. Run python3 tests/verify.py --check behavior, substituting synthesis, defects, benchmark or portability as needed. Each invocation generates fresh retained evidence under evidence/l1/.

The complete reduced benchmark ledger is evidence/l1/l1-benchmark-20261031T134253Z-ef6ca00d/children/verifier-five-domain-matrix-20261031T134253Z-261960b9/benchmark.json; it records precision, cycles, stalls, transactions, storage and matching parent results.

Generate bridge recipes with python3 tests/hardware_vectors.py --generate --format fp16 --weights 0 --cache 1 --banks 4 --seed 17. Compare recipe and trace files with python3 tests/hardware_vectors.py --compare RECIPE TRACE. Generation is not hardware execution.

Use design/anvil_q2_fp.sdc for the 10.000 ns core clock. Supply actual external-interface delay and bridge-clock constraints separately; the measured virtual-pin compilation does not establish those interfaces. Device addressing and cables remain external configuration.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 22 Verilog/SystemVerilog files; 3237 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/anvil_q2_bank_scheduler.sv	233 total	227 code
rtl/anvil_q2_bus_mux.sv	35 total	35 code
rtl/anvil_q2_fp_00_pkg.sv	380 total	340 code
rtl/anvil_q2_fp_add.sv	112 total	107 code
rtl/anvil_q2_fp_awq_dequant.sv	68 total	68 code
rtl/anvil_q2_fp_cache_codec.sv	138 total	138 code
rtl/anvil_q2_fp_decoder_layer.sv	219 total	219 code
rtl/anvil_q2_fp_dot_lane.sv	78 total	77 code
rtl/anvil_q2_fp_elementwise.sv	134 total	134 code
rtl/anvil_q2_fp_exp.sv	101 total	97 code
rtl/anvil_q2_fp_gemv.sv	106 total	106 code
rtl/anvil_q2_fp_gemv_bank.sv	248 total	248 code
rtl/anvil_q2_fp_gqa_attention.sv	252 total	250 code
rtl/anvil_q2_fp_kv_cache.sv	257 total	255 code
rtl/anvil_q2_fp_parameters.sv	36 total	36 code
rtl/anvil_q2_fp_rmsnorm.sv	148 total	148 code
rtl/anvil_q2_fp_rope.sv	145 total	145 code
rtl/anvil_q2_fp_scalar.sv	132 total	129 code
rtl/anvil_q2_fp_swiglu_mlp.sv	91 total	91 code
rtl/anvil_q2_fp_top.sv	277 total	275 code
rtl/anvil_q2_mem_port.sv	49 total	49 code
rtl/anvil_q2_scalar_arbiter.sv	63 total	63 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md

Reference	Original project-relative record
E003	design/SPEC.md
E015	rtl/anvil_q2_fp_gemv.sv
E027	tests/attention_checks.py
E040	tests/fp_reference.py
E043	tests/lane_lifecycle_checks.py
E060	tests/suite_manifest.py
E066	tests/transformer_reference.py
E1158	.vulcan/engineering/run-0001/attempt-1175/stdout.log
E1159	.vulcan/engineering/run-0001/attempt-1176/stdout.log
E1160	.vulcan/engineering/run-0001/attempt-1177/stdout.log
E1161	.vulcan/engineering/run-0001/attempt-1178/stdout.log
E1162	.vulcan/engineering/run-0001/attempt-1179/stdout.log
E1191	.vulcan/engineering/run-0001/rtl-timing-lint.json
E1192	.vulcan/engineering/run-0001/quartus-sta.json
E1193	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).